Retail Store Sales & Profitability Analysis

Excel Analytics Project Report

Tool: Microsoft Excel
Project Type: Retail Sales & Business Analytics
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1. Project Overview

This project analyzes retail store sales and transaction data to understand sales performance, profitability, orders, products, delivery performance, returns, geographic performance and payment behavior. Microsoft Excel was used for data validation, data preparation, calculated fields, KPI analysis, descriptive statistics, business analysis and dashboard development.

2. Business Understanding

Business Problem

The business has transactional sales data, but the raw data does not clearly show how sales, profitability, products, delivery performance and returns are performing. The project aims to convert the available data into clear business information that can support better decisions.

Business Objectives

- Analyze sales revenue, cost and profitability.
- Identify high- and low-performing products and categories.
- Analyze completed and returned orders.
- Evaluate delivery performance.
- Understand geographic and payment patterns.
- Create meaningful KPIs and an Excel dashboard.
- Translate findings into actionable business recommendations.

Key Business Questions

- What are the total sales, cost and profit?
- Which product categories perform best?
- Which categories and products are most profitable?
- What is the order return rate?
- How long do orders take to deliver?
- Which countries generate the most revenue?
- What business areas require improvement?

3. Data Understanding

The analysis is based on the 'Retail Store Sales' table containing 555 records and 18 columns.

The table contains information about:

- Orders and customers
- Products and categories
- Order and delivery dates
- Quantity and unit price
- Order status
- Country
- Payment method
- Calculated business metrics

Main Fields

Field	Data Type

Order ID	Numeric
Customer Name	Text/Categorical
Product Category	Text/Categorical
Product Name	Text/Categorical
Order Date	Date
Delivered Date	Date
Quantity	Numeric
Unit Price	Numeric
Status	Text/Categorical
Country	Text/Categorical
Payment Method	Text/Categorical
Year	Numeric
Month	Text/Categorical
Day	Text/Categorical
Delivery time	Numeric
Total Cost	Numeric
Sales Revenue	Numeric
Net Profit	Numeric

4. Data Preparation & Validation

The data was reviewed before analysis to improve reliability and ensure that the calculations and visualizations were based on consistent information.

Validation Checks

- Checked missing values.
- Checked duplicate Order IDs.
- Reviewed data types and date fields.
- Checked numerical fields for valid values.
- Reviewed categorical fields such as status, category, country and payment method.

Calculated Fields Used

- Year
- Month
- Day
- Delivery time
- Total Cost
- Sales Revenue
- Net Profit

5. KPI Analysis

KPI	Result
Sales Revenue	1,471,555.77
Total Cost	957,254
Net Profit	514,301.77
Profit Margin	34.95%
Total Transactions	555
Returned Orders	268
Return Rate	48.3%
Average Delivery Time	7.85 days

The KPI results indicate that the business is profitable overall, with an estimated 34.95% profit margin. However, the return rate of approximately 48.3% is a major area requiring further investigation.

6. Business & Statistical Analysis

6.1 Product Category Performance

Category	Revenue	Profit	Approx. Margin
Books	320,050	144,186	45.05%
Groceries	293,726	132,073	44.97%
Apparel	323,605	90,596	28.00%
Electronics	291,366	76,261	26.18%
Home Decor	242,809	71,186	29.32%

Apparel generates the highest category revenue, but Books generate the highest profit. This demonstrates that the category with the highest sales is not necessarily the most profitable.

6.2 Return & Delivery Analysis

Order Status	Average Delivery Time
Completed	~6.98 days
Returned	~8.77 days

Returned orders take approximately 1.79 additional days to deliver on average. This is an important relationship to investigate, but it should not be interpreted as proof that delivery delays cause returns.

6.3 Descriptive Statistics

Metric	Net Profit Result
Mean	926.67
Median	651
Standard Deviation	838.88
Skewness	1.29

The positive skewness indicates that a smaller number of high-profit transactions influence the average. Therefore, median and distribution should be considered alongside the mean.

6.4 Geographic Performance

Country	Revenue
Nigeria	333,585
Australia	299,488
United Kingdom	277,832
United States	232,807
China	230,231
Brazil	61,582
Antarctica	36,031

Nigeria is the strongest revenue-generating country in the analyzed data. Revenue should be evaluated together with profit, return rate and delivery performance before making investment decisions.

6.5 Monthly Performance

December is the strongest month in the analyzed data, with approximately 156.4K revenue and 55.9K profit. March is also a strong period, while May is among the weaker revenue periods. Strong and weak months can be compared to identify product mix, customer behavior, promotions or seasonal factors.

6.6 Payment Method Analysis

Payment Method	Transactions
Bank Transfer	162
Mobile Money	143
Cash	127
Credit Card	123

Bank Transfer is the most frequently used payment method. Payment behavior can be further compared by country, revenue and return rate.

7. Dashboard

The Excel dashboard consolidates the key performance indicators and major business trends into a single view. It is intended to help management quickly monitor sales, profitability, product/category performance, returns, delivery and geographic patterns.

Dashboard components include KPI summaries, category/product analysis, geographic analysis, trends, order/return analysis and delivery-related views.

8. Key Business Insights

- The business generated approximately 1.47M in revenue and 514.3K in net profit, resulting in an overall profit margin of about 34.95%.
- The return rate is approximately 48.3%, making returns the most important business risk identified.
- Books and Groceries have approximately 45% profit margins and represent strong profitability opportunities.
- Apparel generates the highest category revenue but has a lower margin of about 28%, showing that high sales do not necessarily mean high profitability.
- Electronics has an approximate 26.2% margin and requires a closer product-level profitability review.
- Returned orders take approximately 1.79 days longer to deliver than completed orders.
- Nigeria is the strongest revenue-generating country in the analyzed data.
- December is the strongest month for revenue and profit.

9. Business Recommendations

Reduce returns: Investigate return causes by product, category, country and delivery time, with priority given to high-return products.

Investigate delivery delays: Compare delivery-time groups with return rates and review logistics performance by country and product.

Prioritize high-margin categories: Protect and selectively grow Books and Groceries because of their strong profitability.

Improve low-margin categories: Review pricing, costs, discounts and returns in Apparel and Electronics before pursuing additional volume.

Study strong markets: Investigate why Nigeria performs strongly and compare revenue with profitability, returns and delivery performance.

Use strong periods as benchmarks: Study December and other strong months to understand the factors behind better performance and apply useful lessons to weaker periods.

Monitor distributions, not only averages: Use mean, median and distribution analysis when evaluating transaction profitability because a smaller number of high-profit orders influence the average.

10. Priority Action Plan

Priority	Action	Expected Benefit
1	Investigate and reduce the high return rate.	Lower operational cost and revenue leakage.
2	Analyze delivery delays against returns.	Identify potential logistics-related return drivers.
3	Protect high-margin categories.	Increase profitable revenue.

4	Review low-margin categories.	Improve category profitability.
5	Learn from strong markets and periods.	Improve future sales and planning decisions.

11. Conclusion

The analysis shows a profitable retail business with an overall profit margin of approximately 34.95%, but the high return rate is the most important issue requiring further investigation. Books and Groceries are strong profitability categories, while Apparel and Electronics require closer review of margins, pricing and costs. Delivery performance should be analyzed alongside returns, and strong markets and periods should be studied to understand the drivers of successful performance. Overall, the project demonstrates how Excel can transform transactional data into business insights and actionable recommendations.

12. Tools & Skills Demonstrated

Tool: Microsoft Excel

- Data understanding and validation
- Data cleaning and preparation
- Excel formulas and calculated columns
- Pivot Tables and Pivot Charts
- KPI analysis
- Descriptive statistical analysis
- Business analysis
- Data visualization and dashboard development
- Business insights and recommendations

13. Supporting Documentation

The complete project documentation can be supported by the following files:

- Business Understanding
- Data Dictionary
- Business Insights & Recommendations
- Final Excel Workbook
- Dashboard

End of Project Report